Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Junior software developer with 2+ years building **secure Python/C++ applications**, REST APIs, SQL and MongoDB-backed services, JSON/XML data pipelines, and Web UIs; delivered tested integrations handling 2M+ records and 99%+ reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant data platform** with normalized relational schemas and row-level isolation for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexed query and access-pattern optimization.
- Built **ETL-style integrations** across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5,000+ monthly jobs with incremental loads, reconciliation, validation, error handling, and CI/CD releases achieving 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **React Web UIs and Flask REST APIs** for secure uploads, grading, and history workflows, supporting 150+ users and 10,000+ daily requests while maintaining stable production performance.
- Implemented **secure PDF, DOCX, and ZIP ingestion** with validation, concurrent processing, persistent results, and fault-tolerant API execution, processing 1,000+ files per batch while reducing feedback turnaround 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python data and evaluation pipelines** processing 100K+ trajectory records, creating reproducible test scenarios and diagnostics that improved long-horizon planning 18% and strengthened analysis of complex model behavior.
- Documented **model constraints, benchmark results, and failure patterns** in technical reports, translating complex experimental behavior into actionable engineering guidance that improved closed-loop planning stability by 15% across simulations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed **OOP API and ETL components** using AWS Glue, S3, SQL, JSON, CSV, and Parquet transformations, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Shipped **cross-platform integrations through Agile sprints** with Microsoft and GitHub stakeholders, maintaining 20+ unit/integration tests and Jenkins CI/CD across Windows, Linux, and macOS release environments with reliable delivery.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **React Web UIs and Node.js REST APIs** across eight microservices for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through WebSockets and MongoDB across locations.
- Implemented **role-based access, JWT authentication, automated testing**, and Docker deployment workflows through Jenkins CI/CD, cutting release time 50% while strengthening secure delivery and reducing environment drift across staged environments.

Skills

Languages: Python, C++, JavaScript, TypeScript, Java, SQL, HTML, CSS
Data & Formats: MongoDB, PostgreSQL, MySQL, SQL Server, Azure SQL, JSON, XML, YAML, NoSQL
Web & APIs: React.js, Next.js, Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets
Security & QA: RLS, RBAC, JWT, data validation, secure ingestion, error handling, PyTest, Jest, Selenium
Delivery: Git, GitHub, Agile/Scrum, CI/CD, Jenkins, Docker, AWS, Azure, unit testing, integration testing

Projects

HMIS Live | *TypeScript, MERN, ABDM APIs, Sentry*
- Built a **secure healthcare information platform** with consent-based record access, appointment workflows, TypeScript dashboards, ABDM API integrations, Twilio notifications, and Sentry monitoring, reducing administrative workload by 30% for staff.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered database workspace** supporting 10+ SQL and NoSQL systems, accelerating query authoring, schema exploration, and data management through a Java and TypeScript codebase with Docker-backed development workflows.

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **secure MongoDB-backed platform** supporting 50,000+ users and 1M+ transactions, combining token-based access, REST integrations, real-time updates, filtering, and Kubernetes-based CI/CD automation for scalable operations.